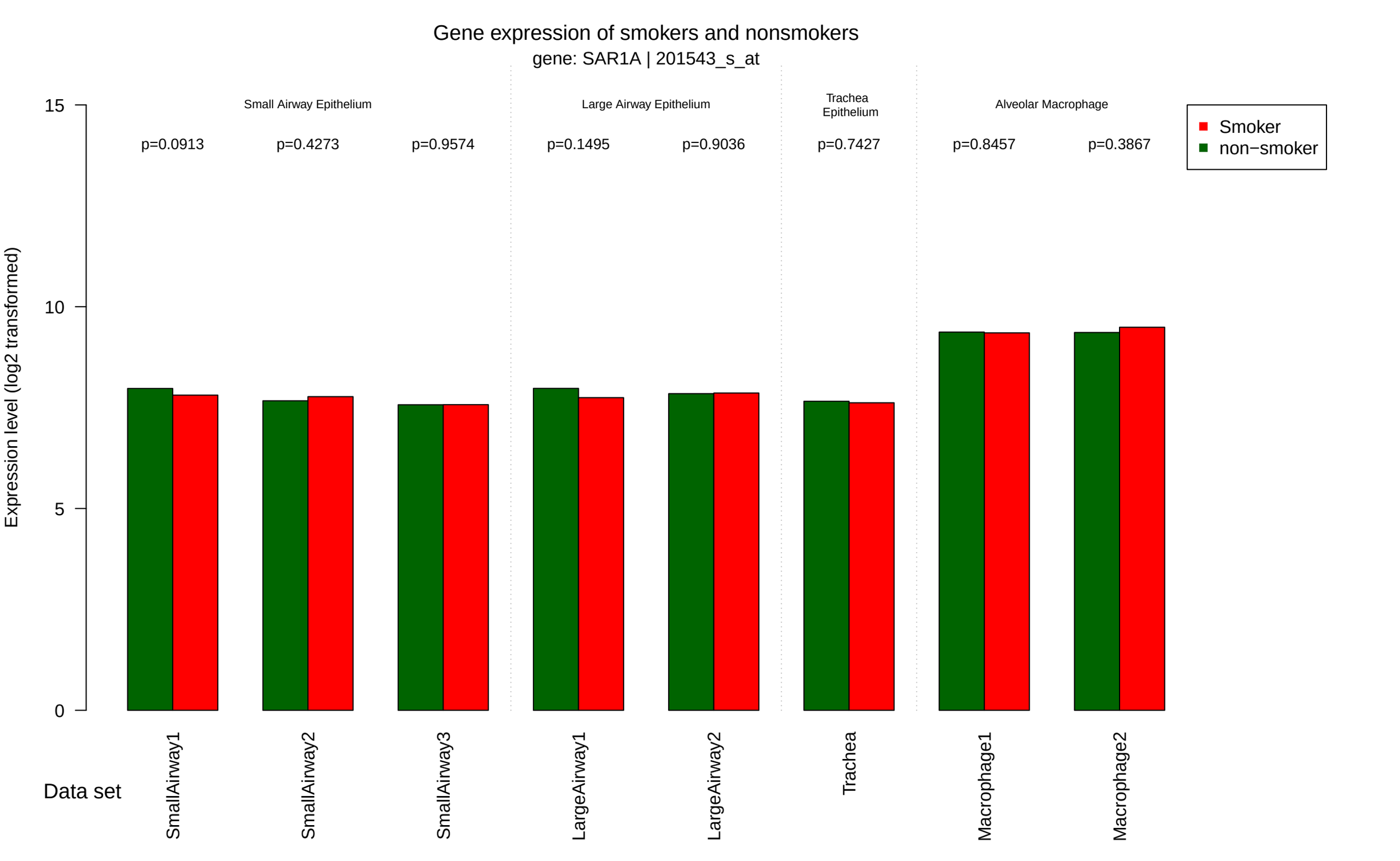
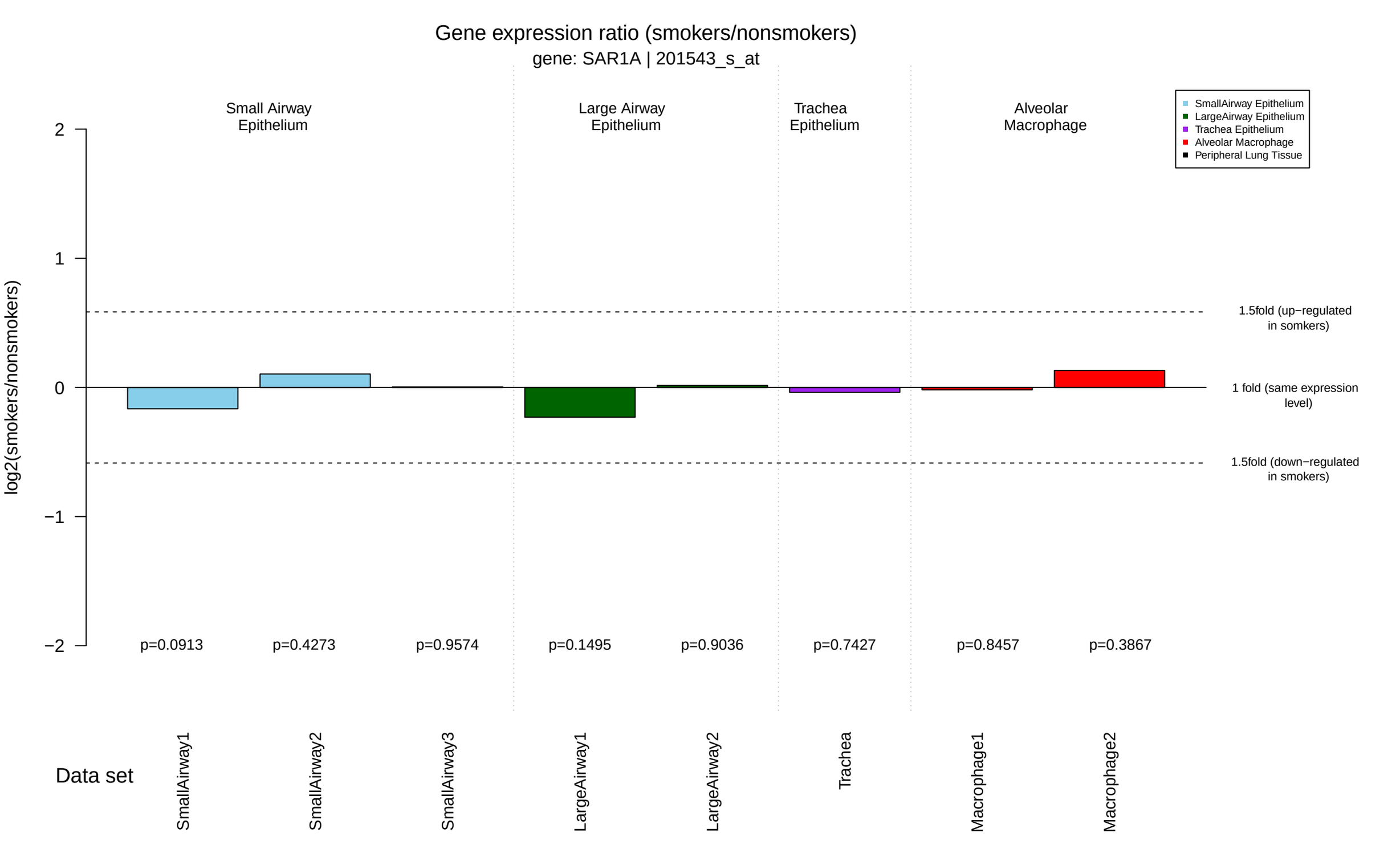
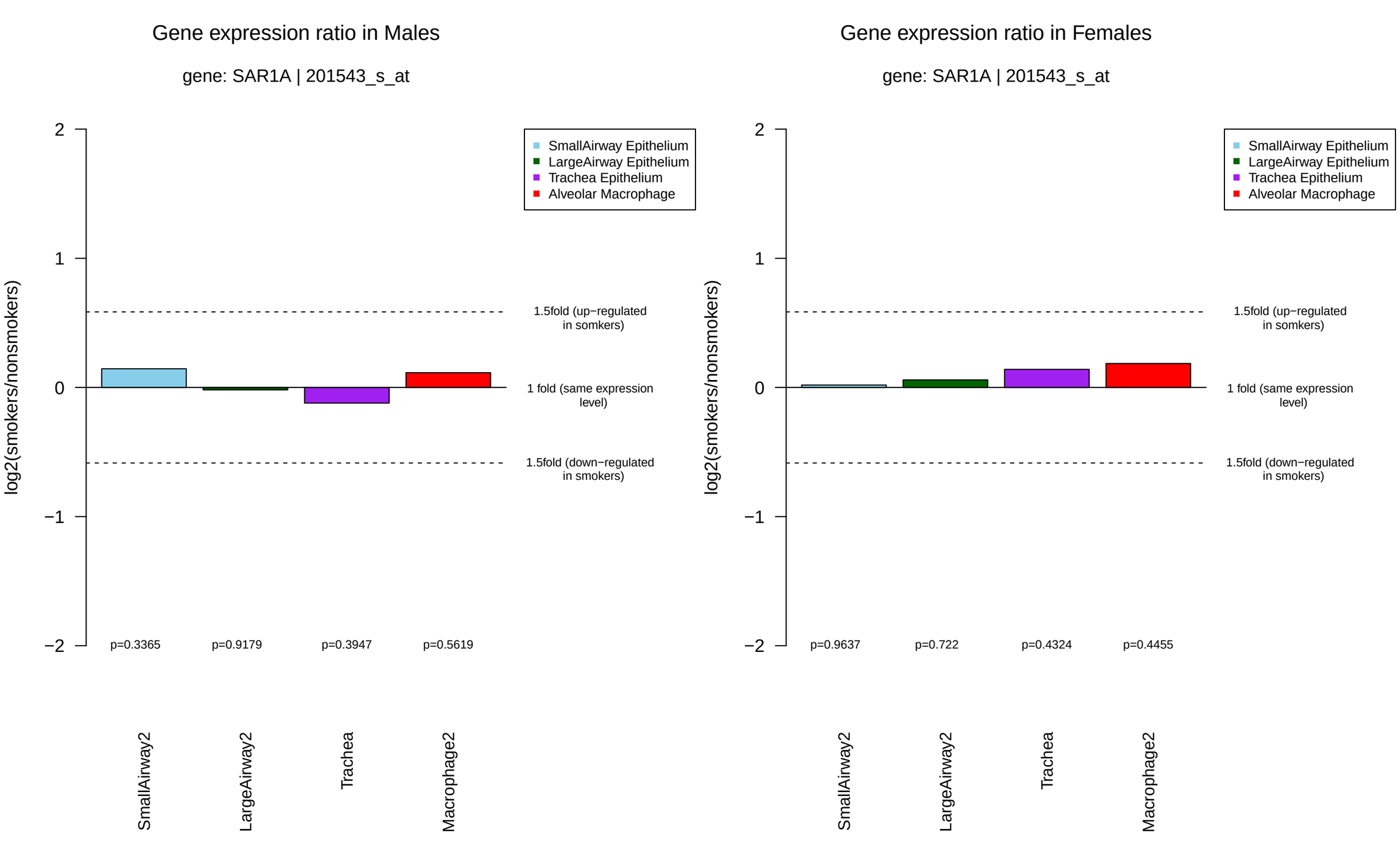
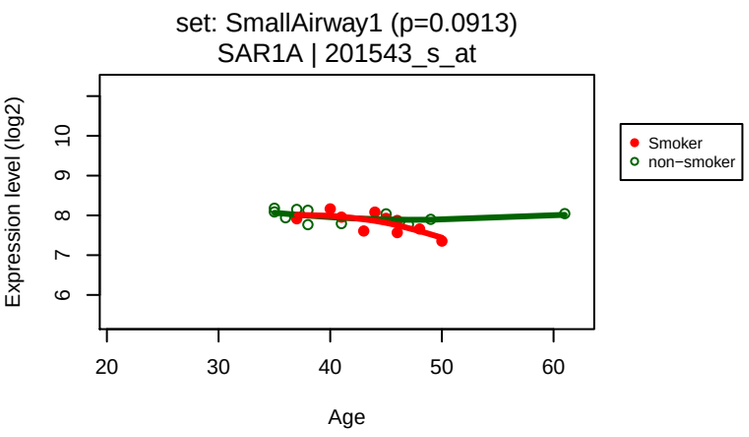
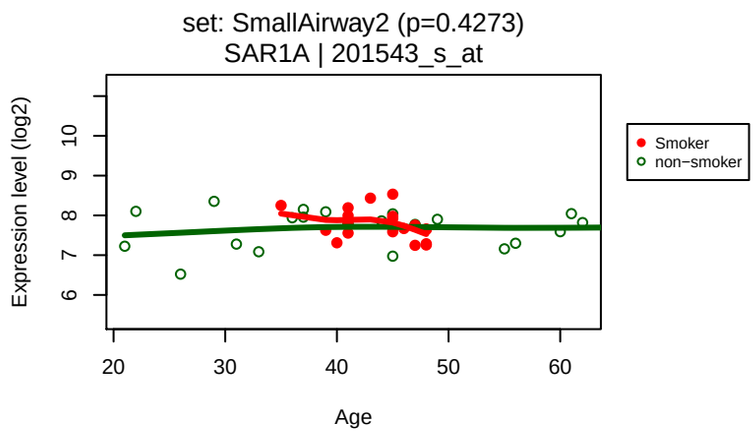
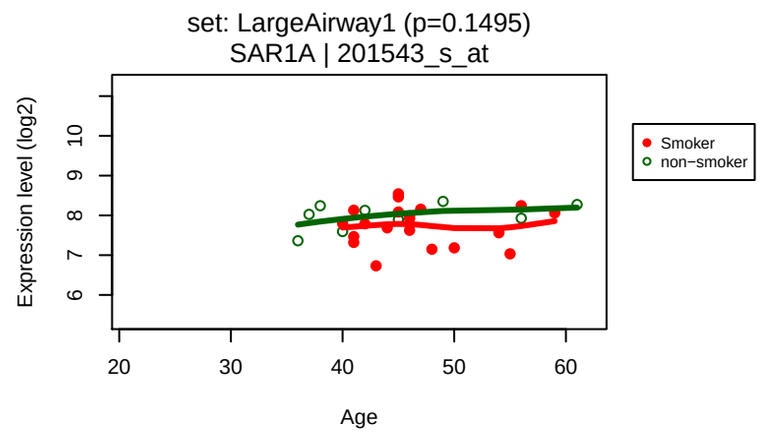

Gene expression of smokers and nonsmokers
gene: SAR1A | 201543_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0913
p=0.4273
p=0.9574
p=0.1495
p=0.9036
p=0.7427
p=0.8457
p=0.3867
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

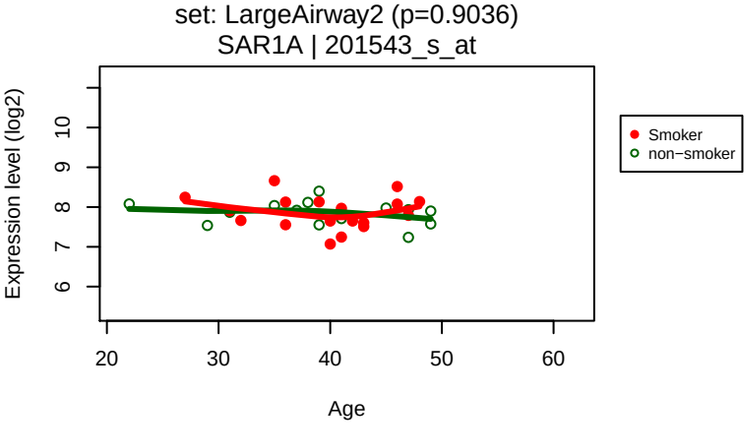

Gene expression ratio (smokers/nonsmokers)
gene: SAR1A | 201543_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0913
p=0.4273
p=0.9574
p=0.1495
p=0.9036
p=0.7427
p=0.8457
p=0.3867
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

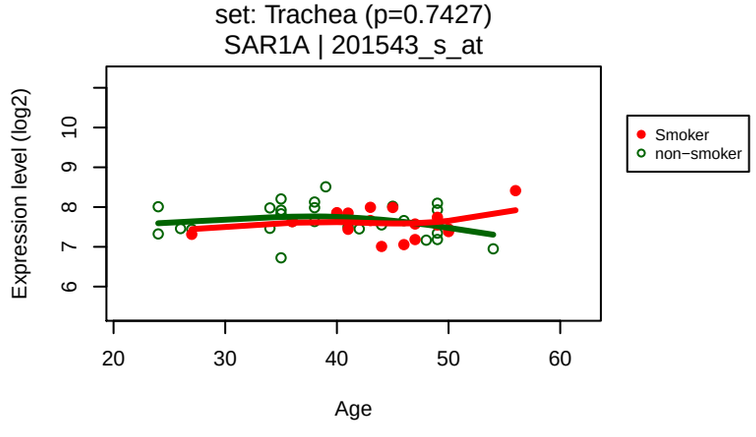

Gene expression ratio in Males
gene: SAR1A | 201543_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3365
p=0.9179
p=0.3947
p=0.5619
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SAR1A | 201543_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9637
p=0.722
p=0.4324
p=0.4455
SmallAirway2
LargeAirway2
Trachea
Macrophage2

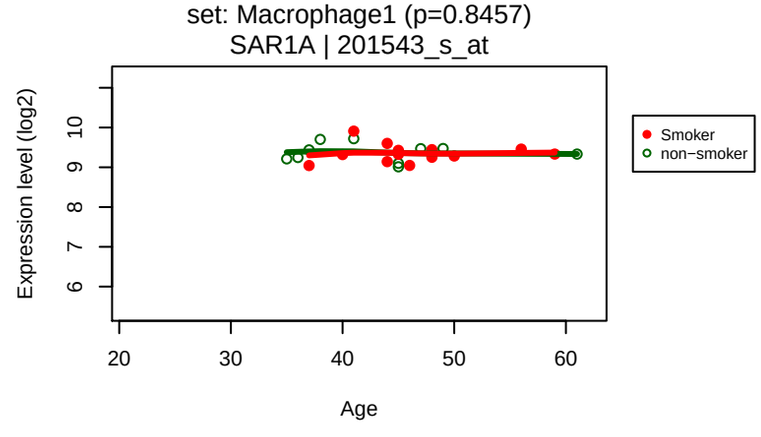

set: SmallAirway1 (p=0.0913)
SAR1A | 201543_s_at
Expression level (log2)
Age
Smoker
non−smoker

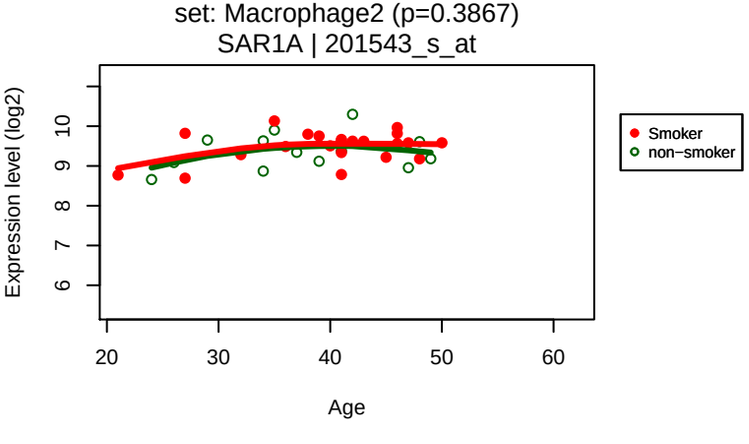

set: SmallAirway2 (p=0.4273)
SAR1A | 201543_s_at
Expression level (log2)
Age
Smoker
non−smoker

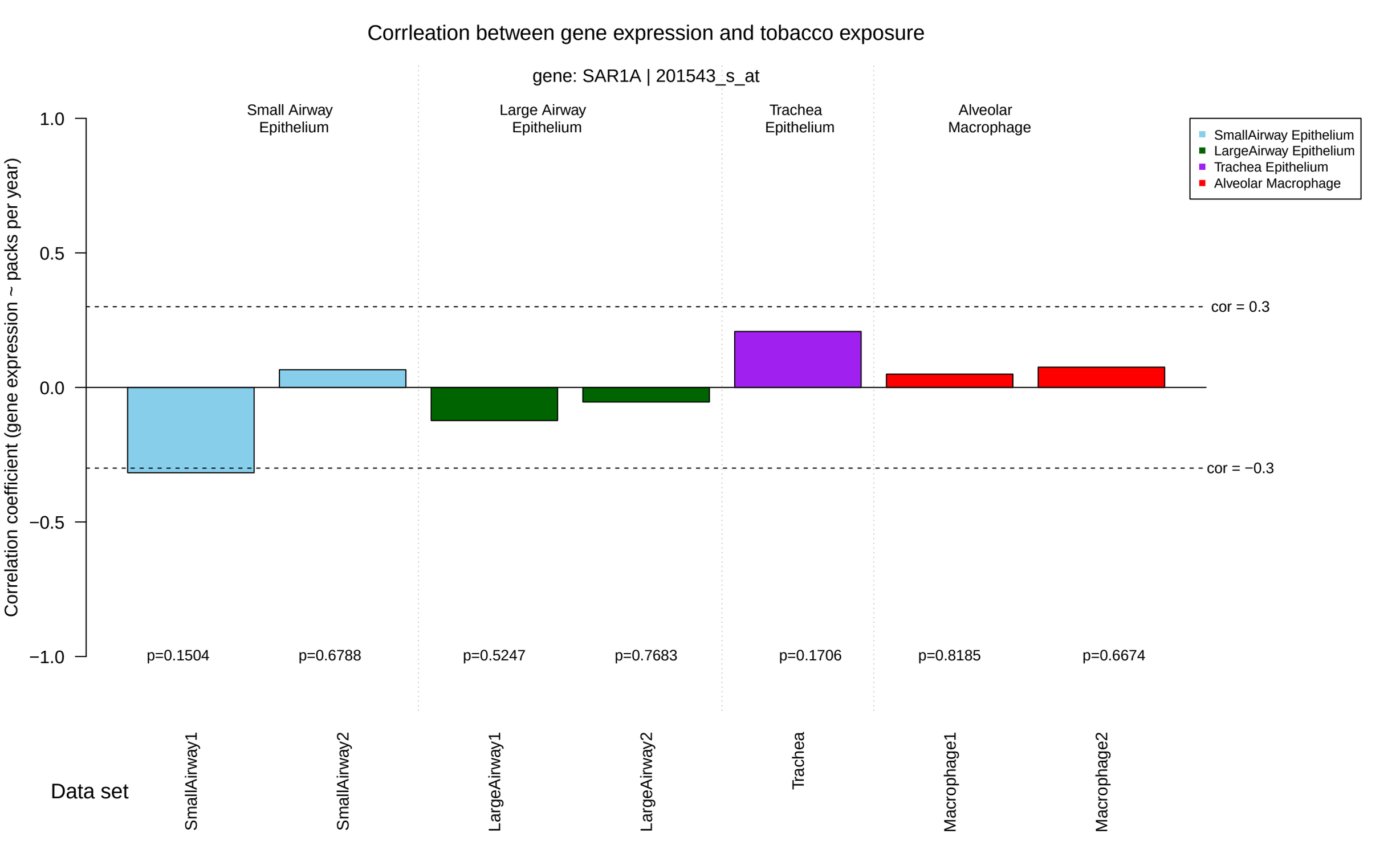

set: LargeAirway1 (p=0.1495)
SAR1A | 201543_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9036)
SAR1A | 201543_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7427)
SAR1A | 201543_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8457)
SAR1A | 201543_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3867)
SAR1A | 201543_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SAR1A | 201543_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1504
p=0.6788
p=0.5247
p=0.7683
p=0.1706
p=0.8185
p=0.6674
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2